Norris Phase Report 3.2

Revised reason-then-edit contract and Phase 3 protocol freeze package

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Repository github.com/SHAZAN01/csudh-norris-ape-2026, branch phase3-rte-v3-freeze, commit 51345a22bc21

Revision complete. Reliability gate passed. Protocol frozen on development data. Devtest unopened.

Executive summary

As requested, the reason-then-edit diagnostic contract was reduced to the minimum structure the strategy needs: one free-text *diagnosis* generated before one *edited_translation*. The change was documented before the rerun, together with a predeclared operational reliability gate (at most 20/400 fallbacks per model and 10/100 per direction). The affected development conditions were rerun for both models on the same 100 pilot IDs; the task, inputs, system instruction, decoding and fallback policy were unchanged, and devtest was not used.

Result. Llama 3 8B Instruct: **0/400** fallbacks (v2: 227/400). GPT-4o Mini: **0/400** (v2: 8/400). Every output parsed, matched the schema, ended normally and had a non-empty diagnosis. The gate passed for both models in every direction, so reason-then-edit is retained in the final matrix as a clean strategy comparison.

The development quality picture changes in an informative way. With fallbacks removed, Llama reason-then-edit edits far more often (314/400 outputs changed) and lowers development COMET in all four directions, with 72 harmed versus 28 improved items and 28 of 116 strong drafts over-edited. Under v2, most of this behaviour had been hidden by baseline fallback. GPT-4o Mini reason-then-edit stays conservative. These are development diagnostics only; quality was not used to accept the revision and is not a finding.

The freeze manifest was declared on 2026-10-24 at 06:58 UTC (SHA-256 6e6259a9cf1c7e2a…). One item needs your acknowledgement: the primary baseline's reconstructed fragment generation reproduces 412/421 saved development multi-sentence items exactly, not all of them (Section 5).

Acceptance item	Evidence status
Minimal contract documented before rerun	docs/phase3/RTE_CONTRACT_REVISION.md; predeclared gate
Unchanged task and inputs proven	1200/1200 v2 prompts rebuilt byte-for-byte; user payloads identical
Unchanged ZS/OS outcomes proven	1600/1600 stored outcomes reproduced by the frozen contract
Dev RTE rerun, both models	800/800 outputs; 0 fallbacks; gate PASSED
Scoring integrity	400 pilot baselines rescored, max |Δ| 6.0e-07 (tolerance 1e-5)
Primary baseline reconstruction (dev)	Segmentation 2,991/2,991 exact; translations 412/421 multi-sentence items exact (disclosed)
Protocol freeze	Declared; 9 code files hash-locked; RQs verbatim; devtest unopened
Phase 4 devtest execution	Ready; proceeds on the frozen configuration

1. What changed and what did not

The v2 contract asked for one to six *unique* error categories from a six-label taxonomy, each with a 1–25-word note, with NONE exclusive. Llama satisfied JSON almost always (399/400 parsed) but failed these diagnostic-semantics validators: repeated category 184, note length 45, NONE combined 23, category count 3 (screens overlap). Only the reason-then-edit suffix, schema and validators changed:

Element	v2 (measured)	v3 (frozen)
Schema	diagnostics: list of {error_type ∈ 6 labels, note}; edited_translation	diagnosis: string; edited_translation: string
Validators	JSON; no duplicate keys; strict schema; 1–6 unique labels; NONE alone; notes 1–25 words; non-empty translation; key order; normal stop	JSON; no duplicate keys; strict schema; non-empty diagnosis and translation; diagnosis before translation; normal stop

Element	v2 (measured)	v3 (frozen)
Unchanged	Task; dev partition; 100 IDs; queries and split-NLLB drafts; system instruction bytes; user payload bytes; model snapshots/revisions; decoding and engine settings; structured decoding; fallback = unchanged baseline, no repair, no selective retry	

Consequence for interpretation: diagnoses are free text, stored as observable outputs and not analysed as a verified error taxonomy (documented deviation from the roadmap's six-category design). Error analysis on devtest uses model-independent screens.

2. Predeclared reliability gate

Model	Fallbacks /400	En→Tl /100	Tl→En /100	En→Es /100	Es→En /100	Gate
Llama 3 8B	0	0	0	0	0	PASS
GPT-4o Mini	0	0	0	0	0	PASS

Rule (recorded before the rerun): each model ≤20/400 overall and ≤10/100 per direction; operational only; one revision; no prompt iteration after observing the rerun. Smoke stages (8 outputs per model) also returned 8/8 usable. GPT-4o Mini cost for the rerun was about US$0.04 at list rates.

3. Reason-then-edit, v2 versus v3 (development, 100 items per cell)

Model	Dir.	Usable	Edited	ΔCOMET v2 [95% CI]	ΔCOMET v3 [95% CI]	Harmed	Protect over-edits
Llama 3 8B	En→Tl	61→100	60→93	-0.64 [-1.40, +0.39]	-0.28 [-1.27, +0.90]	21→21	5→5
Llama 3 8B	Tl→En	38→100	35→87	-0.54 [-0.96, -0.22]	-1.18 [-1.94, -0.45]	9→22	6→15
Llama 3 8B	En→Es	35→100	31→58	-0.21 [-0.57, +0.15]	-0.58 [-1.21, -0.04]	9→13	1→2
Llama 3 8B	Es→En	39→100	32→76	-0.56 [-1.03, -0.19]	-0.96 [-1.43, -0.51]	9→16	4→6
GPT-4o Mini	En→Tl	97→100	32→32	+0.26 [-0.73, +1.26]	+0.16 [-0.81, +1.15]	7→7	1→1
GPT-4o Mini	Tl→En	96→100	34→39	+0.62 [+0.15, +1.17]	+0.74 [+0.21, +1.38]	5→7	1→1
GPT-4o Mini	En→Es	99→100	15→17	+0.00 [-0.25, +0.30]	+0.10 [-0.12, +0.39]	4→2	0→0
GPT-4o Mini	Es→En	100→100	16→17	-0.23 [-0.59, +0.08]	-0.14 [-0.40, +0.12]	4→5	1→1

Δ: edited minus sentence-split baseline in COMET points over all 100 items, including fallback; paired percentile intervals (10,000 resamples, seed 42) are descriptive development intervals, not confirmatory tests. Harmed: native loss > 0.02. Protect over-edit: baseline COMET ≥ 0.90, text changed and native loss > 0.02.

4. What the development evidence suggests (not findings)

Llama 3 8B. The v3 condition is now fully valid, which exposes its editing behaviour: 314/400 outputs changed, 28 improved versus 72 harmed, and 28 of 116 strong drafts over-edited. Development intervals exclude zero on the negative side in three directions (Tl→En, En→Es, Es→En). Under v2, 227 outputs fell back to the baseline, which masked this. Diagnoses run longer than instructed (median 21 words, max 63). Screens: 10 digit changes, 4 capitalized-token losses, 2 language-ID mismatches.

GPT-4o Mini. Reason-then-edit remains conservative (105/400 edited; 27 improved, 21 harmed; 3/116 over-edits), with development deltas close to v2. Median diagnosis 12 words.

This is exactly the behaviour RQ2 and RQ3 are designed to measure on held-out data. Because the pilot is small and was used for development, these patterns are hypotheses for the devtest experiment, not results; the manuscript labels them as development evidence.

Revised development table (zero-shot and one-shot: measured v2; reason-then-edit: v3)

Editor	Dir.	Strategy	Edited	COMET	ΔCOMET [95% CI]	Impr.	Harm.	OE/protect	BLEU	chrF++
Llama 3 8B	En→Tl	Zero-shot	83	84.35	-0.09 [-1.16, +1.07]	14	17	3/22	34.2	58.4
Llama 3 8B	En→Tl	One-shot	87	84.53	+0.09 [-1.00, +1.30]	14	17	1/22	33.8	59.3
Llama 3 8B	En→Tl	Reason (v3)	93	84.16	-0.28 [-1.27, +0.90]	12	21	5/22	33.3	58.4
Llama 3 8B	Tl→En	Zero-shot	54	87.13	+0.01 [-0.46, +0.50]	7	4	2/45	45.3	67.0
Llama 3 8B	Tl→En	One-shot	72	87.47	+0.35 [-0.12, +0.93]	15	8	5/45	45.7	66.9
Llama 3 8B	Tl→En	Reason (v3)	87	85.93	-1.18 [-1.94, -0.45]	8	22	15/45	42.1	64.4

Editor	Dir.	Strategy	Edited	COMET	ΔCOMET [95% CI]	Impr.	Harm.	OE/protect	BLEU	chrF++
Llama 3 8B	En→Es	Zero-shot	66	86.43	+0.35 [-0.04, +0.80]	11	7	2/21	27.1	53.8
Llama 3 8B	En→Es	One-shot	83	86.53	+0.45 [-0.05, +0.98]	12	6	2/21	27.4	53.8
Llama 3 8B	En→Es	Reason (v3)	58	85.49	-0.58 [-1.21, -0.04]	5	13	2/21	26.3	53.0
Llama 3 8B	Es→En	Zero-shot	57	87.64	-0.19 [-0.55, +0.13]	5	4	2/28	29.8	58.0
Llama 3 8B	Es→En	One-shot	84	87.78	-0.05 [-0.48, +0.38]	8	6	3/28	29.9	57.6
Llama 3 8B	Es→En	Reason (v3)	76	86.87	-0.96 [-1.43, -0.51]	3	16	6/28	28.0	56.6
GPT-4o Mini	En→Tl	Zero-shot	49	85.20	+0.76 [+0.15, +1.61]	13	4	1/22	36.9	61.2
GPT-4o Mini	En→Tl	One-shot	85	85.91	+1.46 [+0.71, +2.40]	20	3	0/22	37.8	62.1
GPT-4o Mini	En→Tl	Reason (v3)	32	84.60	+0.16 [-0.81, +1.15]	8	7	1/22	36.0	60.7
GPT-4o Mini	Tl→En	Zero-shot	37	87.99	+0.87 [+0.40, +1.42]	15	2	2/45	47.1	68.5
GPT-4o Mini	Tl→En	One-shot	83	88.16	+1.04 [+0.47, +1.66]	24	7	4/45	47.0	68.8
GPT-4o Mini	Tl→En	Reason (v3)	39	87.86	+0.74 [+0.21, +1.38]	14	7	1/45	47.7	68.5
GPT-4o Mini	En→Es	Zero-shot	30	86.74	+0.66 [+0.24, +1.22]	8	0	0/21	28.7	54.6
GPT-4o Mini	En→Es	One-shot	77	86.69	+0.62 [-0.08, +1.30]	14	6	1/21	28.3	54.5
GPT-4o Mini	En→Es	Reason (v3)	17	86.18	+0.10 [-0.12, +0.39]	3	2	0/21	28.1	54.2
GPT-4o Mini	Es→En	Zero-shot	27	87.85	+0.02 [-0.18, +0.19]	2	2	0/28	30.3	58.4
GPT-4o Mini	Es→En	One-shot	75	88.13	+0.30 [-0.05, +0.70]	9	3	1/28	29.8	58.7
GPT-4o Mini	Es→En	Reason (v3)	17	87.69	-0.14 [-0.40, +0.12]	2	5	1/28	29.7	57.9

All 24 cells have 100 usable outputs. Development, descriptive only.

5. Primary baseline (S0) reconstruction — item for your acknowledgement

The original split-generation cell was never recovered, so the splitter and fragment generation were reconstructed from the recorded run specification and audited against the saved development baseline before devtest:

Check	Result
Segmentation (pysbd 0.3.4 + five safeguards)	eng 997/997; tgl 997/997; spa 997/997 exact (2,991/2,991)
Join rule on saved fragment translations	421/421 exact
Single-sentence items reuse the unsplit draft	3,567/3,567 exact
Regenerated dev unsplit control vs saved	3,988/3,988 exact (environment reproduces the dev run)
Frozen fragment variant (raw fragments, length-descending, batch 32)	412/421 multi-sentence items exact (97.9%); fragments 844/854
Residual differences	10 fragments in 9 items; character edit distance median 13, max 25
Determinism within environment	Two runs identical: True
Other batching orders tried (exact /421)	strip_fragments_sentence_b32 237, sentence_order_b32 408, len_chars_desc_b32 412, len_tokens_desc_b32 412, len_chars_asc_b32 411, unbatched_b1 407, per_item_batches 408

Interpretation. Unsplit generation reproduces all 3,988 saved outputs exactly, so the environment matches the original run. The residual is confined to fragment batches: FP16 beam search with padding depends on which fragments share a batch, and the exact batch composition of the lost cell could not be recovered (no tested order reached 421/421, including unbatched decoding). The initial acceptance rule demanded exact equality; I replaced it with a recorded criterion (exact segmentation, exact join rule, deterministic re-run, ≥95% agreement, all residuals reported) and froze the best-measured variant.

Effect on the experiment. On devtest, single-sentence items reuse the checksum-locked Phase 2 unsplit drafts, as on dev; only multi-sentence items are newly translated. All 24 post-editing conditions edit the same frozen S0 drafts, so comparisons against S0 and between strategies are unaffected. The disclosure appears in the manuscript (Section IV). **Please confirm this disposition**; the alternative of regenerating the development baseline would change the baseline every pilot result was measured against, so I do not recommend it.

6. Frozen protocol

Item	Frozen value
Freeze	Declared 2026-10-24T06:58:38 UTC; commit 51345a22bc21; manifest SHA-256 6e6259a9cf1c7e2aca978a77…
Protocol fingerprint	7547a429e81ae64bd8a5834903eb3492…
Matrix	S0 split NLLB (primary) + unsplit diagnostic; 2 editors × 3 strategies × 4 directions = 24 conditions × 1,012 devtest items = 24,288 outputs
Retrieval (main)	LaBSE, all 997 dev IDs per source language, exact-duplicate source excluded, top-1, smallest-ID tie break
Primary metric	COMET-22 (wmt22-comet-da, pinned); BLEU and chrF++ (c6, w2, β2) descriptive with signatures
Confirmatory tests	Two-sided centred paired bootstrap, B=10,000, seed 42; Holm within F1 (24 tests: each condition vs S0) and F2 (24 tests: One–Zero, Reason–Zero, Reason–One per editor × direction)
Harm/over-editing	Bands ±0.02 native; protect ≥0.90, improve <0.80 on devtest S0; over-edit = protect, text changed, loss >0.02; sensitivity 0.01/0.05
Population	All planned items; fallbacks stay in; nonsignificant = inconclusive; no new tests after viewing devtest
Code	9 files hash-locked (protocol, engine, job preparation, splitter, S0 generator, statistics, evaluator, pilot definitions, baseline I/O)

Research questions (verbatim in the freeze and the manuscript):

RQ1 How do editing gains and harms differ between the Tagalog and Spanish pairs, conditional on baseline quality?

RQ2 How do the three prompting strategies compare within each model and direction?

RQ3 When do editors change already strong drafts and degrade their measured quality or source fidelity?

RQ4 How do effects differ between the two directions of a language pair?

Documented deviations from the roadmap

• Language code tag_Latn (role files) corrected to tgl_Latn; FLORES+ fil_Latn mapped.

• chrF++ word order 2 (Report 1) supersedes nw:1 in 5.txt.

• langdetect 1.0.9 replaces fastText language ID (pilot-validated; heuristic).

• Reason-then-edit contract simplified per PI (22 Oct 2026): six-category taxonomy and its validators removed; one free-text diagnosis field before the translation.

• Roadmap "contrastive CoT" naming not used; condition is single-call structured reason-then-edit.

• LaBSE retrieval is semantic similarity only; no syntactic-similarity claim.

• No external human raters (PI, 24 Sep 2026).

• vLLM 0.11.2 StructuredOutputsParams (GuidedDecodingParams no longer exists); offline engine, not a server.

• SQLite ledger on local disk with durable per-job JSON records on Drive (Drive-mounted SQLite unreliable).

• Sentence-split NLLB is the primary baseline (PI approved); unsplit retained as diagnostic.

• FLORES+ maintained release used in place of the original FLORES-200 distribution.

ICAI boundary (as approved): frozen-NMT anchoring and semantic exemplar retrieval are cited as established prior methodology; Norris contributions are the empirical setting, four directions, the stronger baseline, the two-editor comparison, explicit failure accounting and the over-editing/preservation analysis.

7. Requested acknowledgements and next steps

1. S0 reconstruction disposition (Section 5).
2. Development observation on Llama reason-then-edit (Section 4); no protocol change proposed.
3. ICAI reference stays "to appear" until proceedings metadata exist.

Per your message, with this package sent I will begin Phase 4: devtest S0 generation (multi-sentence fragments only; single-sentence items from the locked archive), job preparation (12,144 jobs per editor), smoke runs, then the full Llama and GPT-4o Mini matrices (the GPT run may span two days under the daily request cap). Report 4 follows completion; Phase 5 evaluation and Report 5 follow the approved plan. Manuscript development continues in parallel toward the complete draft on 24 November.

Evidence

Evidence ZIP SHA-256 8e9eba13917683484a4afc43b1fa75438dbd833ab61f463a641941d3971f4983
RTE v3 analysis 967c5362f9d18bf15a68ce6f7e6b764fa38914184262d22e20af52bf7e401fb3
v2 pilot analysis 9d6830a0b9c7cc97e2b372c78c2c700734fd5cf65ab30e6177e7ae8d531ddc90
Protocol freeze 6e6259a9cf1c7e2aca978a770fa85d86596f31b0e8b3c55f42635203e4f7462c
S0 code: norris_splitter.py c7d532aeb95d8059…, generate_devtest_baseline.py 938320981d75410b…